QFX5202 Portfolio Analytics - Problem Set 3

Instructions
Work individually and show all steps. Submit a single PDF by Tuesday, 20 October 2026, 6:00 PM.

Question 1 (30 marks): Estimate the covariance matrix of the five assets in the data file and comment on its stability over time.
Question 2 (40 marks): Construct the minimum-variance portfolio with and without short-selling constraints.
Question 3 (30 marks): Compare the out-of-sample Sharpe ratios of both portfolios.